Phylum Brachiopoda

Brachiopods

Occurrence c.400 spp. worldwide; attached to hard surfaces or buried in soft sediments on the seabed

Encased in bivalved shells, usually with a stalk (pedicle) that may anchor the animal to a hard surface or be used in burrowing, brachiopods look remarkably like bivalve mollusks with stalks attached, but the 2 groups are not related. Also known as lamp shells, brachiopods are especially abundant in colder waters. They are less than 4 in (10 cm) long, and most of the space in the shell is taken up by a loop or spiral of hollow, fringed tentacles (lophophore) surrounding the mouth, with which the animal suspension feeds. Brachiopods are extremely common as fossils: some rocks are made up entirely of brachiopod shells. Once an important group comprising more than 25,000 species, brachiopods appear to have declined as a result of competition with bivalve mollusks.

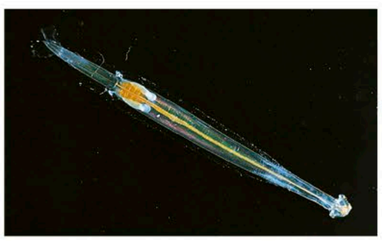
LIOTHYRELLA UVA
Although brachiopods resemble bivalves, the 2 halves of the shell are dorsal (upper) and ventral (lower), not left and right. The species featured, Liothyrella uva, *is common in Atlantic waters, and has a shell that is up to ¾ in (2 cm) long.*

Phylum Chaetognatha

Arrow worms

Occurrence c.150 spp. worldwide; in plankton, one genus on the seabed

Growing only up to 4¾ in (12 cm) long, arrow worms are relatively small, highly active predators that can eat a third of their own body weight a day. They catch their prey—animal plankton, including larval fishes—with movable spines around their mouths and kill it by injecting tetrodotoxin, a potent poison produced by symbiotic bacteria, and used by very few animals.



SAGITTID SP.
The small, torpedo-shaped, transparent arrow worms are carnivores. Like the Sagittid *species above, they live among the plankton in all the oceans—where their large numbers can have great ecological impact; for instance, on fish larvae.*

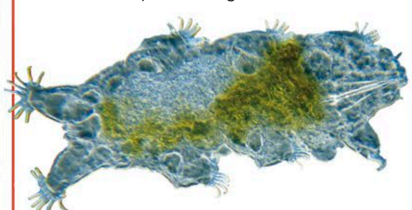
Phylum Tardigrada

Water bears

Occurrence c.1,000 spp. worldwide; in freshwater, salt water, and damp and wet terrestrial habitats

Habitat All except dry ones

Water bears—or tardigrades—are plump-bodied, microscopic animals with 4 pairs of stubby, clawed legs. Their lumbering movements are remarkably bearlike, but their closest relatives are probably velvet worms. They live in damp habitats such as moss, but can shrivel into a dry husk during drought and survive this way for years. Water bears lay eggs, but some females are parthenogenetic.



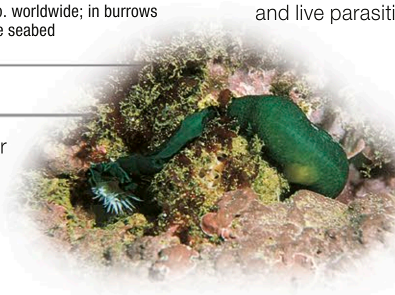
UNIDENTIFIED SP.
Water bears inhabit water films on land and live between sand grains and crevices on the seabed. Many live in clumps of moss on roofs and in gutters. By being dormant for extended periods, they may live over 50 years.

Phylum Echiura

Spoonworms

Occurrence c.200 spp. worldwide; in burrows or other cavities on the seabed

Spoonworms, or echiurans, are sedentary, marine animals with bulbous, sausagelike bodies. Females are up to 20 in (50 cm) long, with an even longer proboscis for sweeping up food. Males are minute and live parasitically inside their partners.



BONELLIA VIRIDIS
This European species has green, toxic skin. Females are up to 6 in (15 cm) long, with a proboscis that can reach out for food over about 3¼ ft (1 m).

Phylum Phoronida

Horseshoe worms

Occurrence c.20 spp. worldwide; in soft sediments or attached to hard surfaces on the seabed

These sedentary marine worms are generally less than 8 in (20 cm) long, and usually remain encased in a secreted chitinous tube, with a horseshoe-shaped crown of as many as 15,000 feeding tentacles around the mouth. The sexes are separate in some species, while others are hermaphrodite. Horseshoe worms are most closely related to brachiopods, or lamp shells (see left).

PHORONIS SP.
Members of the widespread genus Phoronis *can be 10 in (25 cm) long, but most are much smaller. They have an unsegmented body, usually encased in a chitinous tube buried in sand or mud. The body wall has layers of longitudinal and circular muscles, which enable the worm to move up the tube to feed, and to retreat afterward.*

Phylum Onychophora

Velvet worms

Occurrence c.180 spp. in tropical and southern temperate zones; in moist forests, leaf litter, under stones

These are the only survivors of a group of worms that were abundant in the Cambrian Period. They have cylindrical, wormlike bodies, up to 6 in (15 cm) long, 14–43 pairs of short, unjointed, fleshy legs, each terminating in a pair of claws, and an antennae-bearing head with jaws similar in form to the claws on the legs. These carnivorous animals catch their prey by spraying them with a mucuslike substance from a pair of slime glands opening on either side of the mouth; they use the same technique to defend themselves from their predators.

MACROPERIPATUS TORQUATUS
Velvet worms, such as Macroperipatus torquatus *(below), inhabit warm and humid regions. Their anatomy is basically that of a worm but, like arthropods, they have pairs of legs serially repeated along the body. The legs are fleshy, unjointed, and hydraulically operated.*

Phylum Hemichordata

Hemichordates

Occurrence c.130 spp. worldwide; in marine mud and sand, and other intertidal and subtidal habitats

Hemichordates are unusual among invertebrates in that they have some vertebrate characteristics: a nerve cord that runs along the back, and pharyngeal perforations—structures that have the same anatomical origin as fish gills. The length of the 3-part bodies—comprising a proboscis, a collar, and a long trunk—ranges from less than ½ in (1.2 cm) to 8¼ ft (2.5 m). These animals are found in 2 forms: sedentary, polyplike pterobranchs, whose saclike trunk ends in a stalk that may be prehensile or connect individuals to form colonies; and free living, wormlike forms, in which the trunk is stalkless and bears a terminal anus. The sexes are separate, and reproduction occurs either sexually or asexually, by fragmentation or budding.

ACORN WORMS
Slow-moving and fragile, acorn worms live in U-shaped burrows in intertidal and subtidal habitats in all parts of the world. The large trunk of Balanoglossus australiensis *(above), which may be 3¼ ft (1 m) long, bears numerous gill slits.*